

ECE 3113 - Energy Conversion

Homework 5

(Due on Monday 11/20/2023)

1. For a motor with the following characteristics, $R_a = 0.01 \, \Omega$, $V_{dc} = 120 \, \text{V}$, $V_{brush} = 0 \, \text{V}$, no-load speed $n_0 = 1000$ turns/minute,
 - a) Derive the equation of the output curve (i.e., the n-Torque curve).
 - b) Draw the associated Speed-Torque output curve.
2. In the equivalent circuit of DC motor below, $R_a = 0.02 \, \Omega$, $V_{dc} = 120 \, \text{V}$, $V_{brush} = 0 \, \text{V}$, and no-load speed $n_0 = 1800$ turns/min.
 - a) What are maximum output torque T_{max} and maximum current $I_{am,max}$?
 - b) Find equation to represent the output characteristic of DC motor and then draw the output characteristic in the plot of n versus T_{out} .
 - c) What is the efficiency of this motor when $n = 1600$ turns/min?

